eso-pic 宏包*

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1 简介

本宏包可方便地在每一页的绝对位置添加图形命令。

2 用法

另请参见示例 LaTeX 文档（eso-*.tex）。

2.1 将 LaTeX 内容添加到页面背景的基本命令

\AddToShipoutPictureBG 作为 \AddToShipoutPictureBG 命令参数的所有图形命令都将被添加到内部宏 \ESO@HookIBG 中。该宏是一个零长度 picture 环境的一部分，其基点位于页面左下角（默认）或左上角（宏包选项 “texcoord”）。该 picture 环境将在每个新页输出时被送出。

\AddToShipoutPictureBG* \AddToShipoutPictureBG* 与 \AddToShipoutPictureBG 类似，但仅对当前页有效。它使用内部宏 \ESO@HookIIBG，其内容将自动清除。

为兼容起见，宏 \AddToShipoutPicture 和 \AddToShipoutPicture* 分别是 \AddToShipoutPictureBG 和 \AddToShipoutPictureBG* 的别名。

\AddToShipoutPictureFG 与 \AddToShipoutPictureBG 类似，但图形命令位于正常文档内容之上。

\AddToShipoutPictureFG* 调用 \ClearShipoutPictureBG 会清除 \ESO@HookIBG 的内容。

\ClearShipoutPictureBG 为兼容起见，宏 \ClearShipoutPicture 是 \ClearShipoutPictureBG 的别名。

\ClearShipoutPictureFG 调用 \ClearShipoutPictureFG 会清除 \ESO@HookIFG 的内容。

\LenToUnit [允许将长度作为图形命令的参数。] 注意，此宏仅为兼容本宏包旧版本而存在。新版 LaTeX 已允许在图形命令中直接使用长度。

\gridSetup \gridSetup[⟨*gridunitname*⟩]{⟨*gridunit*⟩}{⟨*labelfactor*⟩}{⟨*griddelta*⟩}

*本文档对应 eso-pic v3.0e，日期 2026/05/31。

`{`⟨*gridDelta*⟩`}{`⟨*gap*⟩`}`。详情参见实现部分。

\AtPageUpperLeft
\AtPageLowerLeft
\AtPageCenter
\AtTextUpperLeft
\AtTextLowerLeft
\AtTextCenter
\AtStockUpperLeft
\AtStockLowerLeft
\AtStockCenter

辅助宏，用于更方便地在页面上定位。

2.2 宏包选项

选项	取值	说明
pscoord	空或 *true*、*false*	坐标系原点为页面左下角（默认）。
texcoord	空或 *true*、*false*	坐标系原点为页面左上角。
grid	空或 *true*、*false*	是否绘制网格；默认：false。
gridBG	空或 *true*、*false*	将网格置于背景；默认：false。
gridunit	*mm*、*in*、*bp*、*pt*	网格线间距为 gridunit 的倍数。默认：mm。
gridcolor	有效颜色名	主网格线颜色；默认：black。
subgridcolor	有效颜色名	次网格线颜色；默认：black。
subgridstyle	*solid* 或 *dotted*	“dotted” 非常慢！默认：solid。
colorgrid	空或 *true*、*false*	相当于 ‘grid=true’、‘gridcolor=red’ 和 ‘subgridcolor=green’ 的快捷方式；默认：false。
dvips	空或 *true*、*false*	若 color 或 graphics 使用了 dvips 选项则也为 true。为 true 时加载 eepic 宏包以获得更好的点线性能。
enabled	空或 *true*、*false*	控制是否启用本宏包的输出（仅用于特殊情况）；默认：true。

3 实现

3.1 主要功能

\LenToUnit 仅为兼容保留

```
\providecommand*\LenToUnit[1]{#1}

\newdimen\ESO@tempdima
\newdimen\ESO@tempdimb
```

\ESO@isMEMOIR 对 memoir 文档类的支持。

```
\newcommand\ESO@isMEMOIR[2]{#2}
\@ifclassloaded{memoir}{\renewcommand\ESO@isMEMOIR[2]{#1}}{}
```

\AtPageUpperLeft

```
\newcommand\AtPageUpperLeft[1]{%
  \put(0,-\ESO@yoffsetI){#1}%
```

```
}
```

\AtPageLowerLeft

```
\newcommand\AtPageLowerLeft[1]{\AtPageUpperLeft{%
  \put(0,-\paperheight){#1}}}
```

\AtPageCenter

```
\newcommand\AtPageCenter[1]{\AtPageUpperLeft{%
  \put(.5\paperwidth,-.5\paperheight){#1}}%
}
```

\AtStockLowerLeft

```
\newcommand\AtStockLowerLeft{}
\ESO@isMEMOIR{%
  \renewcommand\AtStockLowerLeft[1]{%
    \begingroup
      \ESO@tempdima=-\stockwidth
      \advance\ESO@tempdima\trimedge
      \advance\ESO@tempdima\paperwidth
      \if@twoside\ifodd\c@page\else
        \ESO@tempdima=-\trimedge
      \fi\fi
      \ESO@tempdimb=-\stockheight
      \advance\ESO@tempdimb\paperheight
      \advance\ESO@tempdimb\trimtop
      \AtPageLowerLeft{%
        \put(\ESO@tempdima,\ESO@tempdimb){#1}%
      }%
    \endgroup
  }
}{%
  \let\AtStockLowerLeft=\AtPageLowerLeft
}
```

\AtStockUpperLeft

```
\newcommand\AtStockUpperLeft{}
\ESO@isMEMOIR{%
  \renewcommand\AtStockUpperLeft[1]{%
    \AtStockLowerLeft{%
      \put(0,\stockheight){#1}%
    }%
  }%
}{%
```

```
  \let\AtStockUpperLeft=\AtPageUpperLeft
}
```

\AtStockCenter

```
\newcommand\AtStockCenter{}
\ESO@isMEMOIR{%
  \renewcommand\AtStockCenter[1]{%
    \AtStockLowerLeft{%
      \put(.5\stockwidth,.5\stockheight){#1}%
    }%
  }%
}{%
  \let\AtStockCenter=\AtPageCenter
}
```

\AtTextUpperLeft

```
\ESO@isMEMOIR{%
  \newcommand\AtTextUpperLeft[1]{%
    \begingroup
      \if@twoside
        \ifodd\c@page
          \ESO@tempdima=\spinemargin
        \else
          \ESO@tempdima=\paperwidth
          \advance\ESO@tempdima-\textwidth
          \advance\ESO@tempdima-\spinemargin
        \fi
      \else
        \ESO@tempdima=\spinemargin
      \fi
      \ESO@tempdimb=-\uppermargin
      \AtPageUpperLeft{%
        \put(\ESO@tempdima,\ESO@tempdimb){#1}%
      }%
    \endgroup
  }%
}{%
  \newcommand\AtTextUpperLeft[1]{%
    \begingroup
      \ESO@tempdima=1in\relax
      \if@twoside
        \ifodd\c@page
          \advance\ESO@tempdima\oddsidemargin
```

```
        \else
          \advance\ESO@tempdima\evensidemargin
        \fi
      \else
        \advance\ESO@tempdima\oddsidemargin
      \fi
      \ESO@tempdimb=-1in\relax
      \advance\ESO@tempdimb-\topmargin
      \advance\ESO@tempdimb-\headheight
      \advance\ESO@tempdimb-\headsep
      \AtPageUpperLeft{%
        \put(\ESO@tempdima,\ESO@tempdimb){#1}%
      }%
    \endgroup
  }
}
```

\AtTextLowerLeft

```
\newcommand\AtTextLowerLeft[1]{\AtTextUpperLeft{%
  \put(0,-\textheight){#1}}}
```

\AtTextCenter

```
\newcommand\AtTextCenter[1]{\AtTextUpperLeft{%
  \put(.5\textwidth,-.5\textheight){#1}}}
```

\ESO@HookIBG
\ESO@HookIIBG
\ESO@HookIIIBG
\ESO@HookIFG
\ESO@HookIIFG
\ESO@HookIIIFG

```
\newcommand{\ESO@HookIBG}{}
\newcommand{\ESO@HookIIBG}{}
\newcommand{\ESO@HookIIIBG}{}
\newcommand{\ESO@HookIFG}{}
\newcommand{\ESO@HookIIFG}{}
\newcommand{\ESO@HookIIIFG}{}
\newcommand{\ESO@HookI}{}
\newcommand{\ESO@HookII}{}
\newcommand{\ESO@HookIII}{}
\let\ESO@HookI=\ESO@HookIBG
\let\ESO@HookII=\ESO@HookIIBG
\let\ESO@HookIII=\ESO@HookIIIBG
```

\AddToShipoutPictureBG
\AddToShipoutPictureFG

```
\newcommand{\AddToShipoutPictureBG}{%
  \@ifstar{\g@addto@macro\ESO@HookIIBG}{\g@addto@macro\ESO@HookIBG}}
\newcommand{\AddToShipoutPicture}{}
```

```
117 \let\AddToShipoutPicture=\AddToShipoutPictureBG
```

\AddToShipoutPictureFG

```
118 \newcommand{\AddToShipoutPictureFG}{%
119   \@ifstar{\g@addto@macro\ESO@HookIIFG}{\g@addto@macro\ESO@HookIFG}}
```

\ClearShipoutPictureBG

\ClearShipoutPicture

\ClearShipoutPictureFG

```
120 \newcommand{\ClearShipoutPictureBG}{\global\let\ESO@HookIBG\@empty}
121 \newcommand{\ClearShipoutPicture}{}
122 \let\ClearShipoutPicture=\ClearShipoutPictureBG
123 \newcommand{\ClearShipoutPictureFG}{\global\let\ESO@HookIFG\@empty}
```

3.2 背景网格

```
124 \RequirePackage{keyval}
```

初始化部分宏

```
125 \newif\ifESO@enabled\ESO@enabledtrue
126 \newif\ifESO@dvips\ESO@dvipsfalse \newif\ifESO@grid\ESO@gridfalse
127 \newif\ifESO@texcoord\ESO@texcoordfalse \newif\ifESO@gridBG\ESO@gridBGfalse
128 \newcommand*\ESO@gridunitname{}
129 \newcommand*\ESO@gridunit{}
130 \newcommand*\ESO@labelfactor{}
131 \newcommand*\ESO@griddelta{}\newcommand*\ESO@griddeltaY{}
132 \newcommand*\ESO@gridDelta{}\newcommand*\ESO@gridDeltaY{}
133 \newcommand*\ESO@gridcolor{}
134 \newcommand*\ESO@subgridcolor{}
135 \newcommand*\ESO@subgridstyle{dotted}% ???
136 \newcommand*\ESO@gap{}
137 \newcommand*\ESO@yoffsetI{}\newcommand*\ESO@yoffsetII{}
138 \newcommand*\ESO@gridlines{\thinlines}
139 \newcommand*\ESO@subgridlines{\thinlines}
140 \newcommand*\ESO@hline[1]{\ESO@subgridlines\line(1,0){#1}}
141 \newcommand*\ESO@vline[1]{\ESO@subgridlines\line(0,1){#1}}
142 \newcommand*\ESO@Hline[1]{\ESO@gridlines\line(1,0){#1}}
143 \newcommand*\ESO@Vline[1]{\ESO@gridlines\line(0,1){#1}}
144 \newcommand\ESO@fcolorbox[3]{\fbox{#3}}
145 \newcommand\ESO@color[1]{}
146 \newcommand\ESO@colorbox[2]{%
147   \begingroup
148     \fboxrule=\z@\fbox{#2}%
149   \endgroup
150 }
```

\gridSetup

```
\newcommand\gridSetup[6][]{%
  \edef\ESO@gridunitname{#1}\edef\ESO@gridunit{#2}
  \edef\ESO@labelfactor{#3}\edef\ESO@griddelta{#4}
  \edef\ESO@gridDelta{#5}\edef\ESO@gap{#6}}
```

所有键值选项

```
\define@key{ESO}{enabled}[true]{\csname ESO@enabled#1\endcsname}
\define@key{ESO}{texcoord}[true]{\csname ESO@texcoord#1\endcsname}
\define@key{ESO}{pscoord}[true]{\csname @tempswa#1\endcsname
  \if@tempswa\ESO@texcoordfalse\else\ESO@texcoordtrue\fi}
\define@key{ESO}{dvips}[true]{\csname ESO@dvips#1\endcsname}
\define@key{ESO}{grid}[true]{\csname ESO@grid#1\endcsname
  \setkeys{ESO}{gridcolor=black,subgridcolor=black}}
\define@key{ESO}{gridBG}[true]{\csname ESO@gridBG#1\endcsname}
\define@key{ESO}{colorgrid}[true]{\csname ESO@grid#1\endcsname
  \setkeys{ESO}{gridcolor=red,subgridcolor=green}}
\define@key{ESO}{gridcolor}{\def\ESO@gridcolor{#1}}
\define@key{ESO}{subgridcolor}{\def\ESO@subgridcolor{#1}}
\define@key{ESO}{subgridstyle}{\def\ESO@subgridstyle{#1}}%
\define@key{ESO}{gridunit}{%
  \def\@tempa{#1}
  \def\@tempb{bp}
  \ifx\@tempa\@tempb
    \gridSetup[\@tempa]{1bp}{1}{10}{50}{2}
  \else
    \def\@tempb{pt}
    \ifx\@tempa\@tempb
      \gridSetup[\@tempa]{\p@}{1}{10}{50}{2}
    \else
      \def\@tempb{in}
      \ifx\@tempa\@tempb
        \gridSetup[\@tempa]{.1in}{.1}{2}{10}{.5}
      \else
        \gridSetup[mm]{1mm}{1}{5}{20}{1}
      \fi
    \fi
  \fi
}
\setkeys{ESO}{subgridstyle=solid,pscoord=true,gridunit=mm}

\providecommand*\@classoptionslist{}
```

\ProcessOptionsWithKV 此宏使用 'key=value' 语法处理宏包选项。

```
\def\ProcessOptionsWithKV#1{%
  \let\@tempc\@empty
  \@for\CurrentOption:=\@classoptionslist\do{%
    \@ifundefined{KV@#1@\CurrentOption}%
    {}{\edef\@tempc{\@tempc,\CurrentOption,}}}%
  \edef\@tempc{%
    \noexpand\setkeys{#1}{\@tempc\@ptionlist{\@currname.\@currext}}}%
  \@tempc
  \AtEndOfPackage{\let\@unprocessedoptions\relax}}%

\ProcessOptionsWithKV{ESO}%
```

\ESO@div 将长度 #1 除以 #2 × \ESO@gridunit 再加一。结果存于 \@tempcnta。

```
\newcommand\ESO@div[2]{%
  \ESO@tempdima=#1\relax\ESO@tempdimb=\ESO@gridunit\relax
  \ESO@tempdimb=#2\ESO@tempdimb\divide\ESO@tempdima by \ESO@tempdimb
  \@tempcnta\ESO@tempdima\advance\@tempcnta\@ne}

\@ifundefined{colorbox}{%
  \IfFileExists{xcolor.sty}{\RequirePackage{xcolor}}%
    {\RequirePackage{color}}%
}{}%
\let\ESO@color=\color
\renewcommand\ESO@colorbox[2]{%
  \begingroup
    \fboxsep=.4ex %
    \colorbox{#1}{#2}%
  \endgroup
}
\renewcommand\ESO@fcolorbox[3]{%
  \begingroup
    \fboxsep=.5ex %
    \fcolorbox{#1}{#2}{#3}%
  \endgroup
}

\ESO@isMEMOIR{%
  \newcommand*\ESO@MEMOIR@corr{%
    \ESO@tempdima=-\trimedge
    \advance\ESO@tempdima-\paperwidth
    \advance\ESO@tempdima\stockwidth
    \if@twoside\ifodd\c@page\else
```

```
      \advance\ESO@tempdima2\trimedge
      \advance\ESO@tempdima\paperwidth
      \advance\ESO@tempdima-\stockwidth
    \fi\fi
    \ESO@tempdimb=\ESO@yoffsetI
    \advance\ESO@tempdimb-\trimtop
  }
  \ifESO@enabled
    \AddToHook{shipout/background}{%
      \ESO@MEMOIR@corr
      \put(\ESO@tempdima,\ESO@tempdimb){%
        \ESO@HookIIIBG\ESO@HookIBG\ESO@HookIIBG
        \global\let\ESO@HookIIBG\@empty
      }%
    }%
    \AddToHook{shipout/foreground}{%
      \ESO@MEMOIR@corr
      \put(\ESO@tempdima,\ESO@tempdimb){%
        \ESO@HookIFG\ESO@HookIIFG\ESO@HookIIIFG
        \global\let\ESO@HookIIFG\@empty
      }%
    }%
  \fi
}{%
  \ifESO@enabled
    \AddToHook{shipout/background}{%
      \put(0,\ESO@yoffsetI){%
        \ESO@HookIIIBG\ESO@HookIBG\ESO@HookIIBG
        \global\let\ESO@HookIIBG\@empty
      }%
    }%
    \AddToHook{shipout/foreground}{%
      \put(0,\ESO@yoffsetI){%
        \ESO@HookIFG\ESO@HookIIFG\ESO@HookIIIFG
        \global\let\ESO@HookIIFG\@empty
      }%
    }%
  \fi
}
\AtBeginDocument{%
  \ESO@dvipsfalse
  \@ifundefined{Gin@driver}{}%
```

```
  {%
    \ifx\Gin@driver\@empty\else%
      \filename@parse{\Gin@driver}\def\reserved@a{dvips}%
      \ifx\filename@base\reserved@a\ESO@dvipstrue\fi
    \fi
  }%
  \ifESO@dvips\def\@tempb{eepic}\else\def\@tempb{epic}\fi
  \def\@tempa{dotted}%\def\ESO@gap{6\@wholewidth}%
  \ifx\@tempa\ESO@subgridstyle
    \IfFileExists{\@tempb.sty}%
    {%
      \RequirePackage{\@tempb}
      \renewcommand*\ESO@hline[1]{\ESO@subgridlines\dottedline{\ESO@gap}%
        (0,0)(##1,0)}
      \renewcommand*\ESO@vline[1]{\ESO@subgridlines\dottedline{\ESO@gap}%
        (0,0)(0,##1)}
    }{}
  \else
    \ifx\ESO@gridcolor\ESO@subgridcolor
      \renewcommand*\ESO@gridlines{\thicklines}
    \fi
  \fi
% \changes{v3.0b}{2023/03/23}{More code inside \cmd{\AtBeginDocument}}
  \ifESO@texcoord
    \def\ESO@yoffsetI{\z@}\def\ESO@yoffsetII{-\paperheight}
    \edef\ESO@griddeltaY{-\ESO@griddelta}\edef\ESO@gridDeltaY{-\ESO@gridDelta}
  \else
    \def\ESO@yoffsetI{-\paperheight}\def\ESO@yoffsetII{\z@}
    \edef\ESO@griddeltaY{\ESO@griddelta}\edef\ESO@gridDeltaY{\ESO@gridDelta}
  \fi
}
```

\GridPicture

```
\newcommand\GridPicture{%
  \begingroup
    \setlength\unitlength{\ESO@gridunit}%
    \ESO@color{\ESO@subgridcolor}%
```

— horizontal subgrid lines —

```
    \ESO@div{\paperheight}{\ESO@griddelta}%
    \multiput(0,0)(0,\ESO@griddeltaY){\@tempcnta}%
      {\ESO@hline{\paperwidth}}%
```

— vertical subgrid lines —

```
    \ESO@div{\paperwidth}{\ESO@griddelta}%
    \multiput(0,\ESO@yoffsetII)(\ESO@griddelta,0){\@tempcnta}%
      {\ESO@vline{\paperheight}}%
    \ESO@color{\ESO@gridcolor}%
```

— horizontal grid lines —

```
    \ESO@div{\paperheight}{\ESO@gridDelta}%
    \multiput(0,0)(0,\ESO@gridDeltaY){\@tempcnta}%
      {\ESO@Hline{\paperwidth}}%
```

— vertical grid lines —

```
    \ESO@div{\paperwidth}{\ESO@gridDelta}%
    \multiput(0,\ESO@yoffsetII)(\ESO@gridDelta,0){\@tempcnta}%
      {\ESO@Vline{\paperheight}}%
```

— horizontal numbers —

```
    \fontsize{10}{12}\normalfont%
    \ESO@div{\paperwidth}{\ESO@gridDelta}%
    \multiput(0,\ESO@gridDeltaY)(\ESO@gridDelta,0){\@tempcnta}{%
      \@tempcntb=\@tempcnta\advance\@tempcntb-\@multicnt
      \ifnum\@tempcntb>1\relax
        \multiply\@tempcntb by \ESO@gridDelta\relax
        \ESO@tempdima=\@tempcntb sp\ESO@tempdima=\ESO@labelfactor\ESO@tempdima
        \@tempcntb=\ESO@tempdima%
        \makebox(0,0)[c]{\ESO@colorbox{white}{\the\@tempcntb}}%
      \fi}%
```

— vertical numbers —

```
    \ifx\ESO@gridunitname\@empty\def\@tempa{0}\else\def\@tempa{1}\fi
    \ESO@div{\paperheight}{\ESO@gridDelta}%
    \multiput(\ESO@gridDelta,0)(0,\ESO@gridDeltaY){\@tempcnta}{%
      \@tempcntb=\@tempcnta\advance\@tempcntb-\@multicnt%
      \ifnum\@tempcntb>\@tempa\relax
        \multiply\@tempcntb by \ESO@gridDelta\relax%
        \ESO@tempdima=\@tempcntb sp\ESO@tempdima=\ESO@labelfactor\ESO@tempdima
        \@tempcntb=\ESO@tempdima
        \makebox(0,0)[c]{\ESO@colorbox{white}{\the\@tempcntb}}%
      \fi
    }%
```

— the unit label —

```
    \ifx\ESO@gridunitname\@empty\else
      \thicklines\fboxrule=\@wholewidth
```

```
      \put(\ESO@gridDelta,\ESO@gridDeltaY){\makebox(0,0)[c]{%
        \ESO@fcolorbox{\ESO@gridcolor}{white}{%
          \textbf{\ESO@gridunitname}}}}%
    \fi
    \normalcolor%
  \endgroup
}
```

将网格添加到输出图形中（\ESO@HookIIIFG 或 \ESO@HookIIIBG），独立于用户命令 \AddToShipoutPictureBG 和 \AddToShipoutPictureBG*）。

```
\ifESO@grid
  \ifESO@gridBG
    \g@addto@macro\ESO@HookIIIBG{\GridPicture}%
  \else
    \g@addto@macro\ESO@HookIIIFG{\GridPicture}%
  \fi
\fi
```